

Project Design Phase
Proposed Solution Template

Date	2 October 2026
Team ID	SWTID-2026-6545
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Users often have a fixed budget but spend significant time searching and comparing products or services across different platforms. They may struggle to balance price, quality, preferences, and occasion requirements, especially when planning home interiors, parties, or jewelry purchases. Existing search and recommendation options can leave users with too many choices and require manual comparison.
2	Idea / Solution description	PocketSmart AI is an AI-powered budget and recommendation assistant. Users select a planner such as Home Interior, Party, or Jewelry, then provide their budget, preferences, and relevant requirements. The FastAPI backend processes the request and uses Gemini AI to generate personalized, budget-aware recommendations. Suitable product or service options can be presented with platform/source information. SQLite with SQLAlchemy can be used to store user information, planner data, and recommendation history.
3	Novelty / Uniqueness	PocketSmart AI combines budget awareness, user preferences, multiple planning categories, and AI-based recommendations in one workflow. Instead of requiring users to search and compare many platforms manually, the system interprets the user's context and produces structured recommendations. The same assistant can support different planning needs such as home, party, and jewelry, with the possibility of extending to additional categories.
4	Social Impact / Customer Satisfaction	The solution can reduce the time and effort required for purchase and event planning and help users make decisions according to their available budget. Personalized

		recommendations can reduce information overload and make planning easier for students, families, working professionals, and other budget-conscious users. Clear budget-aware suggestions can also help users avoid unnecessary overspending.
5	Business Model (Revenue Model)	The platform can use a freemium model in which basic planning and recommendation features are available to users, while advanced personalization, additional planner categories, premium planning features, or enhanced recommendation tools can be offered as paid features. Revenue can also be explored through affiliate/referral commissions from participating product or service platforms, subject to their programs and terms.
6	Scalability of the Solution	PocketSmart AI can be extended by adding new planner categories such as travel, electronics, gifts, or food. More product/service platforms and recommendation sources can be integrated over time. The React frontend, FastAPI backend, SQLAlchemy data layer, SQLite database, and AI recommendation layer can be developed as modular components, allowing the system to evolve toward a larger web platform and later migrate to a more scalable database when required.